

High-Level Structure of the 824-Qubit Point-Addition Circuit

Technical note on the ECDSAfail low-qubit track

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Abstract

We describe the reversible point-addition circuit submitted in commit `f4bc8e3` of the ECDSAfail challenge. The circuit adds a classical SECP256K1 point to a quantum point using 824 live qubits and an average of 192,490,070 executed Toffoli gates. Its small width is obtained by a packed, fixed-schedule extended Euclidean algorithm, by borrowing dirty qubits from an otherwise passive coordinate register, and by erasing the affine slope from an output-side quotient witness. The presentation follows the high-level, circuit-oriented organization of Section 2 of Schrottenloher [1], while the text and analysis are specific to the challenge implementation.

1 High-Level Structure

Let p be the 256-bit prime

$$p = 2^{256} - 2^{32} - 977, \quad (1)$$

and let $E(\mathbb{F}_p)$ denote the group of points on

$$E : y^2 = x^3 + 7 \pmod{p}. \quad (2)$$

This is the curve SECP256K1. The ECDSAfail point-addition interface receives a point $P = (P_x, P_y)$ in two 256-qubit registers and a fixed point $Q = (Q_x, Q_y)$ in two classical bit registers. Its target operation is

$$U_Q : |P_x, P_y\rangle |Q_x, Q_y\rangle_{\text{cl}} \mapsto |R_x, R_y\rangle |Q_x, Q_y\rangle_{\text{cl}}, \quad R = P + Q. \quad (3)$$

The suffix “cl” is important: Q_x and Q_y select classically conditioned gates and do not occupy 512 additional qubits. Thus the public interface contributes 512 live qubits.

Low-qubit track. The implementation considered here is the submission `99deca8e`, represented by commit `f4bc8e3` [3]. It reports a peak of 824 live qubits and is tied for the smallest width in the public qubit-only ranking as observed on 22 July 2026. This ranking should not be confused with the main product-score frontier: the dashboard labels this particular submission as rejected rather than promoted [4].

Quantum circuits. We count the maximum number of simultaneously live qubits and the number of executed Toffoli gates. The primitive stream additionally contains X and CX gates, measurement/reset uncomputation operations denoted HMR, and conditional CZ phase corrections. A clean three-controlled NOT marker is lowered to two Toffoli gates, one HMR, and one conditional CZ. Measurement removes a temporary Boolean value; the conditional phase correction is required to remove the associated input-dependent phase. Consequently, resetting a qubit to zero is not by itself a valid uncomputation criterion.

Point addition. For a generic affine addition, put

$$d_x = Q_x - P_x, \quad d_y = Q_y - P_y, \quad \lambda = d_y d_x^{-1} \pmod{p}. \quad (4)$$

The output is then

$$R_x = \lambda^2 - P_x - Q_x, \quad R_y = \lambda(Q_x - R_x) - Q_y. \quad (5)$$

The implementation assumes that the divisions used below are defined. In particular, its contract excludes $P_x = Q_x$ and $Q_x = R_x$. Point doubling, vertical-line addition, and the point at infinity are therefore outside this path.

The central uncomputation identity is

$$\lambda = \frac{Q_y - P_y}{Q_x - P_x} = \frac{R_y + Q_y}{Q_x - R_x} \pmod{p}. \quad (6)$$

The first quotient creates the slope. Once the output-side numerator and denominator have been constructed, the second quotient certifies the same value and permits the slope register to be erased. This avoids retaining the complete history of the point addition.

It is convenient to describe the division component by the maps

$$D_{\text{fwd}} : |a, b, 0\rangle \mapsto |a, b, b/a\rangle, \quad (7)$$

$$D_{\text{cancel}} : |a, b, b/a\rangle \mapsto |a, b, 0\rangle, \quad (8)$$

for nonzero $a \in \mathbb{F}_p$. Each map computes a fixed-schedule extended Euclidean circuit and subsequently reverses it, so that its internal work registers return to zero.

Algorithm 1 gives the logical organization. It uses names for mathematical values; the physical circuit overwrites and reuses the same coordinate wires whenever their former values have become redundant.

Algorithm 1 Reversible generic addition of a classical point Q

Require: $P = (P_x, P_y) \in E(\mathbb{F}_p)$ in quantum registers

Require: $Q = (Q_x, Q_y) \in E(\mathbb{F}_p)$ in classical registers

Require: $P_x \neq Q_x$ and $Q_x \neq R_x$

Ensure: the quantum coordinate registers contain $R = P + Q$

- 1: Append one zero extension bit to each coordinate
 - 2: $d_x \leftarrow Q_x - P_x$; $d_y \leftarrow Q_y - P_y$
 - 3: $\lambda \leftarrow D_{\text{fwd}}(d_x, d_y)$
 - 4: $d_y \leftarrow d_y - \lambda d_x$ $\triangleright d_y = 0$
 - 5: Recover P_x from d_x and the classical value Q_x
 - 6: $R_x \leftarrow \lambda^2 - P_x - Q_x$
 - 7: $d'_x \leftarrow Q_x - R_x$
 - 8: $d'_y \leftarrow \lambda d'_x$ $\triangleright d'_y = R_y + Q_y$
 - 9: $\lambda \leftarrow D_{\text{cancel}}(d'_x, d'_y, \lambda)$
 - 10: $R_y \leftarrow d'_y - Q_y$; recover R_x from d'_x
 - 11: Remove the two zero extension bits
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Packed Euclidean algorithm. The route selected by PAPER2607_COHERENT_EEA=1 dispatches both division calls to the `paper2607_eea` backend. Its structure is based on the space-efficient construction of Luo et al. [2]. Instead of allocating distinct full-size registers for all remainders, quotients, and Bezout coefficients, it packs the live portions of these values into two 259-qubit

registers, `Work1` and `Work2`. Four phase bits and four small length/location registers identify the active subranges. The physical storage is therefore shared by mathematical variables whose lifetimes do not overlap.

One Euclidean traversal consists of 1,616 reversible microsteps. Each step performs, in order, a cyclic pre-shift of `Work2`, an update on the certified remainder window, a location-controlled quotient/sign swap, an update of the quotient boundary, coefficient subtraction and restoring addition, a post-shift, and a phase update. Every fourth step also performs a termination test and a conditional exchange of the packed work roles. The fixed horizon follows from the checked continuant bound

$$\sum_i \text{bitlength}(q_i) \leq 404, \quad 4 \cdot 404 = 1616, \quad (9)$$

where the q_i are the Euclidean quotients. If the Euclidean computation terminates early, the remaining scheduled steps act as reversible padding.

A row-wise certificate restricts the arithmetic in each microstep to active windows of the packed state. Both 259-qubit arrays remain allocated, so this does not lower their width; it primarily avoids Toffoli gates on positions known to be irrelevant. The active implementation reads 36 compressed chunks from `paper2607_exactwidth_data`. The older `paper2607_data` documentation describes a different resource profile and is not authoritative for this commit.

Register count. The Euclidean core uses

$$\begin{aligned} N_{\text{EEA}} &= 259_{\text{Work1}} + 259_{\text{Work2}} + 4_{\text{phase}} + 8_{l_t} + 9_{l_q} + 9_{l_s} + 8_{l_{r'}} + 12_{\text{aux}} \\ &= 568 \end{aligned} \quad (10)$$

live qubits. During division the 257-bit passenger coordinate is canonical, so its high extension bit is known to be zero and can be released. The peak width is consequently

$$\boxed{N_Q = 568 + (257 - 1) = 824.} \quad (11)$$

The primitive stream addresses 578 local indices. Ten of these positions are dirty aliases borrowed from the 256-qubit passenger register and restored exactly; they are not newly allocated qubits. Relative to the 512 quantum input/output wires, the peak therefore contains 312 additional live qubits.

Table 1: Width of the active fixed-schedule division core.

Component	Qubits
<code>Work1</code> , <code>Work2</code>	259 + 259
Phase, iteration, and sign bits	4
$l_t, l_q, l_s, l_{r'}$	8 + 9 + 9 + 8
Clean auxiliary workspace	12
EEA core	568
Passenger coordinate	256
Peak point-addition width	824

Coordinate arithmetic. The two coordinates are temporarily represented on 257 wires so that a high wire may hold carry, borrow, or sign information. Additions and subtractions use ripple-carry circuits specialized to a classical addend whenever Q_x or Q_y is involved. Reduction exploits

$$2^{256} \equiv 2^{32} + 977 \pmod{p}, \quad (12)$$

so that overflow is folded with a sparse constant rather than by general division. The multiply-accumulate operation is organized as a Horner loop; it does not retain a 512-qubit product register. This serial organization is expensive in gates but compatible with the width in (11).

Number of Euclidean traversals. The first division runs the 1,616-step stream forward to obtain the slope and backward to clean its work state. The output-side cancellation repeats the same forward/backward pair. Hence a complete point addition contains four Euclidean traversals. The operation counts for one traversal are given in Table 2. In particular,

$$T_{\text{EEA}} = 4 \cdot 45,453,265 = 181,813,060, \quad (13)$$

which is approximately 94.45% of the reported full-circuit count

$$T_{\text{total}} = 192,490,070. \quad (14)$$

The remaining coordinate arithmetic contributes 10,677,010 executed Toffoli gates. The design is thus an explicit time–space tradeoff: a very large reversible computation is repeated in order to avoid a persistent history register.

Table 2: Primitive operations in one embedded EEA traversal.

Operation	Count
X	25,190,680
CX	23,020,144
Executed CCX	45,453,265
HMR	5,378,204
Conditional CZ	5,378,204

Correctness scope. Three mechanisms restore temporary state: exact reversal of the compressed EEA stream, the alternative quotient in (6), and measurement-based uncomputation with phase repair. Dirty passenger qubits may contain arbitrary values but must be returned to those same values; clean lenders depend on a proved zero invariant, such as the high bit of a canonical field element. The challenge validator checks output values, zero ancillas, phase cleanliness, and forward-followed-by-reverse behavior.

These checks provide strong benchmark evidence, but the circuit should not be read as a complete implementation of the elliptic-curve group law or as a machine-checked proof for every field input. Its contract is generic affine addition, and its low width is specific to an interface in which the added point is classical. Within that scope, its defining resource identity is

$$\boxed{824 = 568 \text{ packed EEA-core qubits} + 256 \text{ passenger-coordinate qubits}}, \quad (15)$$

paid for by four fixed-schedule EEA traversals and about 1.925×10^8 executed Toffoli gates.

References

- [1] A. Schrottenloher, “Optimized Point Addition Circuits for Elliptic Curve Discrete Logarithms,” arXiv:2606.02235, 2026. <https://arxiv.org/abs/2606.02235>.
- [2] H. Luo et al., “Quantum Algorithm for Elliptic Curve Discrete Logarithms with Space-Efficient Point Addition,” arXiv:2607.13816, 2026. <https://arxiv.org/abs/2607.13816>.
- [3] ECDSAfail Challenge, “Submission commit f4bc8e3,” 2026. <https://github.com/ecdsafail/ecdsafail-challenge/commit/f4bc8e30074fedccbea8475c7d54af4bcefe7c92>.
- [4] ECDSAfail Challenge, “Public qubit-only ranking,” accessed 22 July 2026. <https://www.ecdsafail.rocks/>.